

$$V_2(s) = V_1(s) \cdot H(s)$$

$$H(s) = \frac{V_2(s)}{V_1(s)} = \frac{Z_R}{Z_R + Z_C} = \frac{R_s}{R_s + \frac{1}{Cs}} = \frac{1}{1 + \frac{1}{Rcs}} = \frac{1}{1 + \frac{1}{R_s C_s}}$$

b) $i_C = \dot{v}_C \cdot V_R = i_C R = R_C \dot{v}$

$$\dot{v} + \frac{1}{RC} v = 0$$

$$e^{\frac{1}{RC}t} \left(\dot{v} + \frac{1}{RC} v \right) = 0$$

$$\int_0^t \frac{d}{dt} \left(e^{\frac{1}{RC}t} \cdot v \right) dt = \int_0^t 0 dt$$

$$v e^{\frac{1}{RC}t} - v(0) = 0$$

$$v_2(t) = v(0) e^{-\frac{1}{RC}t} = e^{-\frac{1}{RC}t} \quad | \quad v(0) = 1$$

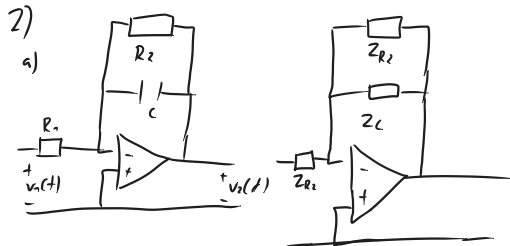
$$V_1(t) = u(t)$$

$$V_2(s) = V_1(s) \cdot H(s)$$

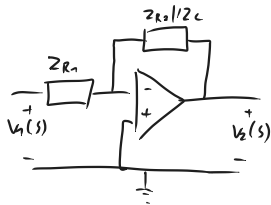
$$= \frac{1}{s} \cdot \frac{1}{1 + \frac{1}{R_s C_s}}$$

$$= \frac{1}{s + \frac{1}{RC}}$$

$$V_2(t) = e^{-\frac{1}{RC}t} = e^{-\frac{1}{RC}t}$$



$$Z_{R2} || Z_C = \frac{1}{\frac{1}{R_2} + Cs} = \frac{R_2}{1 + R_2 Cs} \quad \frac{R_2 \cdot \frac{1}{Cs}}{Cs R_2 + \frac{1}{Cs} \cdot 1}$$



$$V_2(s) = - \frac{Z_{R2} || Z_C}{Z_{R1}} \cdot V_1(s)$$

$$H(s) = \frac{R_2}{1 + R_2 Cs} = - \frac{R_2}{(1 + R_2 Cs) \cdot R_1}$$

$$\Rightarrow - \frac{1}{\left(\frac{1}{R_2} + Cs\right) R_1} = - \left(\frac{1}{R_1 C}\right) \frac{1}{s + \frac{1}{R_2 C}}$$

b) $V_1(t) = u(t)$

$$V_1(s) = \frac{1}{s}$$

$$V_2(s) = V_1(s) \cdot H(s)$$

$$1 \cdot \frac{1}{s} \cdot \frac{1}{1 + \frac{1}{R_1 C s}}$$

$$= \frac{1}{s} \cdot (-1) \cdot (R_1 C) \cdot \frac{1}{s + \frac{1}{R_2 C}}$$

$$= - \left(\frac{1}{s R_1 C \cdot (s + \frac{1}{R_2 C})} \right)$$

$$\Rightarrow - \left(\frac{A}{s R_1 C} + \frac{B}{s + \frac{1}{R_2 C}} \right)$$

Der

$$1 = A \left(s + \frac{1}{R_2 C} \right) + B s R_1 C$$

$$1 = \frac{A}{R_2 C} \Rightarrow A = R_2 C \quad | \quad s = 0$$

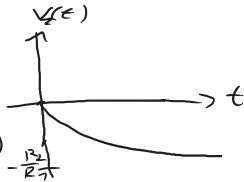
$$1 = - \frac{B R_1 C}{R_2 C} \quad | \quad s = -\frac{1}{R_2 C}$$

$$B = -\frac{R_2}{R_1}$$

$$\Rightarrow - \left(\frac{R_2 C}{s R_1 C} - \frac{R_2}{R_1 \left(s + \frac{1}{R_2 C} \right)} \right)$$

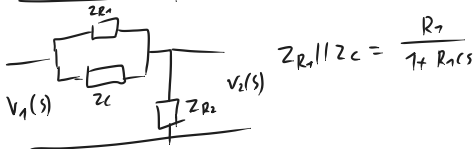
$$\Rightarrow \frac{R_2}{R_1} \left(-\frac{1}{s} + \frac{1}{s + \frac{1}{R_2 C}} \right)$$

$$\Rightarrow \frac{R_2}{R_1} \left(1 - e^{-t/\tau_2} \right) u(t) = v_2(t)$$

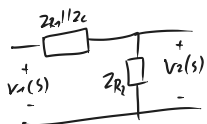


c) kleine Spannung an Probe an setzen
opp diff. Gleichung her, so ble Laplace.

3)



$$Z_{R_1} \parallel Z_C = \frac{R_1}{1 + R_1 C s}$$



$$v_2(s) = H(s) \cdot v_1(s)$$

$$H(s) = \frac{v_2(s)}{v_1(s)} = \frac{Z_{R_2}}{Z_{R_2} + Z_{R_1} \parallel Z_C}$$

$$\Rightarrow \frac{R_2}{R_2 + \frac{R_1}{1 + R_1 C s}} = \frac{R_2 + R_1 R_2 C s}{R_2 + R_1 R_2 (s + R_1)}$$

$$\Rightarrow \frac{\frac{1}{R_1 C} + s}{\frac{1}{R_1 C} + s + \frac{1}{R_2 C}} = \frac{s + \frac{1}{\tau_1}}{s + \frac{1}{\tau_1} + \frac{1}{\tau_2}}$$

$$b) v_1(t) = u(t)$$

$$v_2(s) = v_1(s) \cdot H(s)$$

$$\Rightarrow \frac{1}{s} \cdot \left(\frac{\frac{1}{R_1 C} + s}{s + \frac{1}{R_1 C} + \frac{1}{R_2 C}} \right) = \frac{1}{s(R_1 C)} + \frac{1}{s + \frac{1}{R_1 C} + \frac{1}{R_2 C}}$$

$$\Rightarrow \frac{1}{R_1 C} = A \left(s + \frac{1}{\tau} \right) + B s$$

$$\Rightarrow \frac{1}{R_1 C} = \frac{A}{\tau} \Rightarrow A = \frac{\tau}{R_1 C} \quad | s=0 \quad \tau = \frac{R_1 R_2 C \cdot 1}{(R_1 + R_2) \cdot R_1}$$

$$\Rightarrow \frac{1}{R_1 C} = -\frac{B}{\tau} \Rightarrow B = -\frac{\tau}{R_1 C} \quad | s = -\frac{1}{\tau}$$

$$\Rightarrow \frac{\tau}{R_1 C \cdot s} - \frac{\tau}{R_1 C \cdot (s + \frac{1}{\tau})} + \frac{1}{s + \frac{1}{\tau}} \quad \left| \begin{array}{l} \frac{1}{s R_1 C} \cdot \frac{R_1 R_2 C}{R_1 + R_2} \\ \frac{R_2}{s(R_1 + R_2)} - \frac{R_2}{(R_1 + R_2) \cdot (s + \frac{1}{\tau})} + \frac{1}{s + \frac{1}{\tau}} \end{array} \right|$$

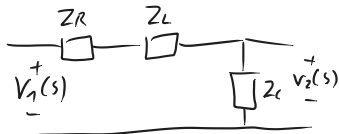
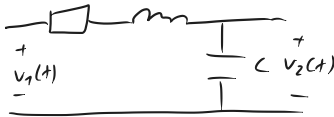
$$\Rightarrow \frac{R_2}{s(R_1 + R_2)} - \frac{R_2}{(R_1 + R_2) \cdot (s + \frac{1}{\tau})} + \frac{1}{s + \frac{1}{\tau}} \quad \left| \begin{array}{l} \frac{R_2}{s R_1 C} \cdot \frac{R_1 R_2 C}{R_1 + R_2} \\ \frac{R_2}{s(R_1 + R_2)} - \frac{R_2}{(R_1 + R_2) \cdot (s + \frac{1}{\tau})} + \frac{1}{s + \frac{1}{\tau}} \end{array} \right|$$

$$\Rightarrow -11 - \frac{-R_2 + R_1 R_2}{(R_1 + R_2) \cdot (s + \frac{1}{\tau})}$$

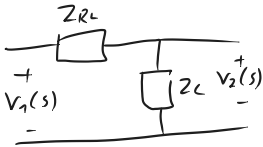
$$\Rightarrow \frac{1}{s} \frac{R_2}{R_1 + R_2} + \frac{R_1}{(R_1 + R_2)} \cdot \frac{1}{s + \frac{1}{\tau}} \quad R_1 s + \frac{R_1}{\tau} + R_2 s + \frac{R_2}{\tau}$$

$$\Rightarrow \left(\frac{R_2}{R_1 + R_2} + \frac{R_1}{R_1 + R_2} e^{-t/\tau} \right) u(t) \quad \begin{array}{c} \frac{R_2}{R_1 + R_2} \\ \frac{R_1}{R_1 + R_2} \end{array}$$

4) a)



$$Z_{RL} = R + Ls \quad Z_C = \frac{1}{Cs}$$



$$H(s) = \frac{Z_C}{Z_C + Z_{RL}} = \frac{\frac{1}{Cs}}{Ls + R + \frac{1}{Cs}}$$

$$\Rightarrow \frac{\frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

6) $\Rightarrow$ Resonanz am für verdringung

$$\frac{1}{LC} = 10^6 \quad \frac{R}{L} = 2000$$

$$\Rightarrow \frac{10^6}{s^2 + 2000s + 10^6} \cdot \frac{1}{s}$$

$$\Rightarrow \frac{10^6}{(s + 1000)^2 + 10^6} \Rightarrow \frac{A}{(s + 1000)^2} + \frac{B}{s}$$

$$\Rightarrow \frac{10^6}{(s + 1000)^2 + 10^6} \Rightarrow \frac{A}{(s + 1000)^2} + \frac{B}{s}$$

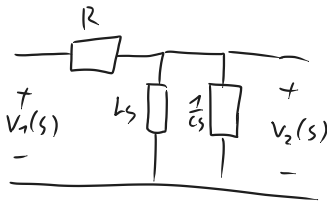
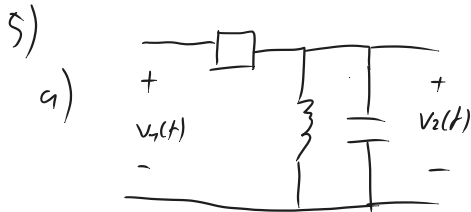
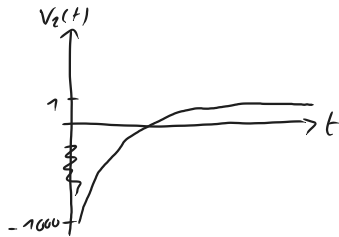
$$\rightarrow 10 = 10^6 \Rightarrow B=1$$

$$\Rightarrow 10^6 = B 10^6 \Rightarrow B=1 \quad | \quad s=0$$

$$\Rightarrow 10^6 = -1000A \Rightarrow A = -1000 \quad | \quad -1000$$

$$\Rightarrow \frac{1}{s} - \frac{1000}{(s+1000)^2}$$

$$\Rightarrow (1 - 1000t e^{-1000t}) u(t)$$



$$V_2(s) = H(s) V_1(s)$$

$$H(s) = \frac{V_2(s)}{V_1(s)} = \frac{Z_L || Z_C}{Z_R + Z_L || Z_C}$$

$$Z_L || Z_C = \frac{Ls \cdot \frac{1}{Cs}}{Ls + \frac{1}{Cs}} = \frac{\frac{L}{C}}{Ls + \frac{1}{Cs}} = \frac{Ls}{LCs^2 + 1}$$

$$\frac{\frac{Ls}{LCs^2 + 1}}{R + \frac{Ls}{LCs^2 + 1}} = \frac{Ls}{RLCs^2 + R + Ls}$$

$$= \frac{s/Rc}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

b)

$$\Rightarrow \frac{s}{s^2 + 1000s + 10^7} \cdot \frac{1}{s}$$

$$\Rightarrow \frac{1000}{s^2 + 1000s + 10^7}$$

$$-1000 \pm \sqrt{(1000)^2 - 4 \cdot 10^7}$$

$$s = \frac{\quad}{2}$$

$$s = -500 \pm 500\sqrt{39}i$$

Komplex Konjugat $p_i = p_i^* \Rightarrow A_j = A_i^*$

$$\frac{1000^7}{(s - 500(1 + \sqrt{39}i))(s - 500(1 - \sqrt{39}i))} = \frac{A_j}{(s - 500(1 + \sqrt{39}i))} + \frac{A_i}{(s - 500(1 - \sqrt{39}i))}$$

$$\frac{1}{1000} = A_j(s - 500(1 - \sqrt{39}i)) + A_i(s - 500(1 + \sqrt{39}i))$$

$$\frac{1}{1000} = A_j(-1000(1 - \sqrt{39}i))$$

$$A_j = -\frac{1}{10^6(1 - \sqrt{39}i)} \Rightarrow A_i = -\frac{1}{10^6(1 + \sqrt{39}i)}$$

$$v_2(t) = -e^{500t} (A_j e^{\sqrt{39}it} + A_i e^{-\sqrt{39}it})$$